PHARMACY INVENTORY & EXPIRY MANAGEMENT SYSTEM

C++ Object-Oriented Architecture & Implementation

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PROBLEM IDENTIFICATION

01

Manual Records
Paper registers lead to errors and stock mismatch

02

Expired Stock
Expired medicines can be sold unknowingly, a direct health risk

03

Slow Search
Finding a medicine or checking its stock takes time

04

Billing Errors
Manual bill calculation is slow and error-prone

KEY OBJECTIVES

Maintain a digital record of all medicines

Detect expired medicines automatically

Block the sale of expired stock

Enable fast search, billing and stock updates

Goal: Safe, accurate and fast pharmacy operations, with zero expired medicines reaching customers

LITERATURE REVIEW

Our system
Expiry Check: Automatic | Blocks Expired Sale: Yes
Lightweight, free, simple

Gap identified: small pharmacies need a simple, low-cost tool that stops expired medicines from being sold.

01

Manual ledger
Expiry Check: No | Blocks Expired Sale: No
Limitation: Error-prone, slow

02

Spreadsheets
Expiry Check: Manual | Blocks Expired Sale: No
Limitation: Needs constant upkeep

03

Commercial pharmacy software
Expiry Check: Yes | Blocks Expired Sale: Yes
Limitation: Costly, complex for small stores

PROPOSED APPROACH

Built on Object-Oriented Programming: two classes, clear responsibilities

01

Model
Medicine class holds each record

02

Store
vector keeps all medicines

03

Interact
Menu-driven console

04

Validate
Every input is checked

05

Protect
Expiry compared with system date

SYSTEM ARCHITECTURE & HIGH-LEVEL DESIGN

01

main() | Entry point
- **Menu display**
- **do-while loop**
- **switch-case calls**

02

Medicine | Data class
- **id, name, quantity**
- **price, day/month/year**
- **Medicine(), display()**

Pharmacy | Manager class

- private: vector<Medicine>
- private: isExpired()
- public: add, display, search, sell, checkExpiry, update, delete

Composition: one Pharmacy object manages many Medicine objects; data stays private to Pharmacy (encapsulation)

TECHNOLOGY STACK & CORE TOOLS

C++ | Language
Fast, compiled, supports OOP

OOP | Concepts
Classes, objects, constructors, encapsulation

STL | vector, string
Dynamic storage and text handling

<ctime> | Library
Reads today's date for expiry checks

Compiler: g++ (any standard C++ compiler)

Interface: Console (terminal)

DATA PIPELINE & SYSTEM WORKFLOW

Overall program loop
Start → Show menu → Read choice → Run function → Repeat until 8 → Exit

Sale pipeline (core logic) →
Any "No" stops the sale with a clear message

01

Enter ID
Medicine found?

02

Not expired?
Valid quantity?

03

Update stock + bill

IMPLEMENTATION: CORE CLASSES & MENU

Expiry check logic

```
if (m.year < cy)
    return true;
if (m.year == cy &&
    m.month < cm)
    return true;
if (m.year == cy &&
    m.month == cm &&
    m.day < cd)
    return true;
return false;
```

L7–35

Medicine class
Stores ID, name, quantity, price, expiry

L15–34

Constructor + display()
Initializes each object; prints its details

L42–56

isExpired()
Gets today's date via ctime; compares year → month → day

L224–282

main()
do-while menu + switch-case; exits on 8

Common pattern: search by ID

```
for (auto &med : medicines) {
  if (med.id == id) {
    // add / show / update
    return;
  }
}
cout << "Medicine not
        found!";
```

IMPLEMENTATION: INVENTORY OPERATIONS

addMedicine() · L60–99
Rejects duplicate IDs, validates input, stores record

displayMedicines() · L101–110
Lists all records or shows empty message

searchMedicine() · L112–125
Linear search by ID, O(n)

updateQuantity() · L181–204
Replaces stock; rejects negative values

IMPLEMENTATION: SALES & EXPIRY

Sale logic (simplified)

```
if (isExpired(med)) return;

if (qty <= 0 ||
    qty > med.quantity)
    return;

med.quantity -= qty;
bill = qty * med.price;
```

01

sellMedicine() · L127–161
Blocks expired stock, checks quantity, prints bill

02

checkExpiry() · L163–179
Scans all medicines, reports expired ones

03

deleteMedicine() · L206–222
Finds index, removes it with erase()

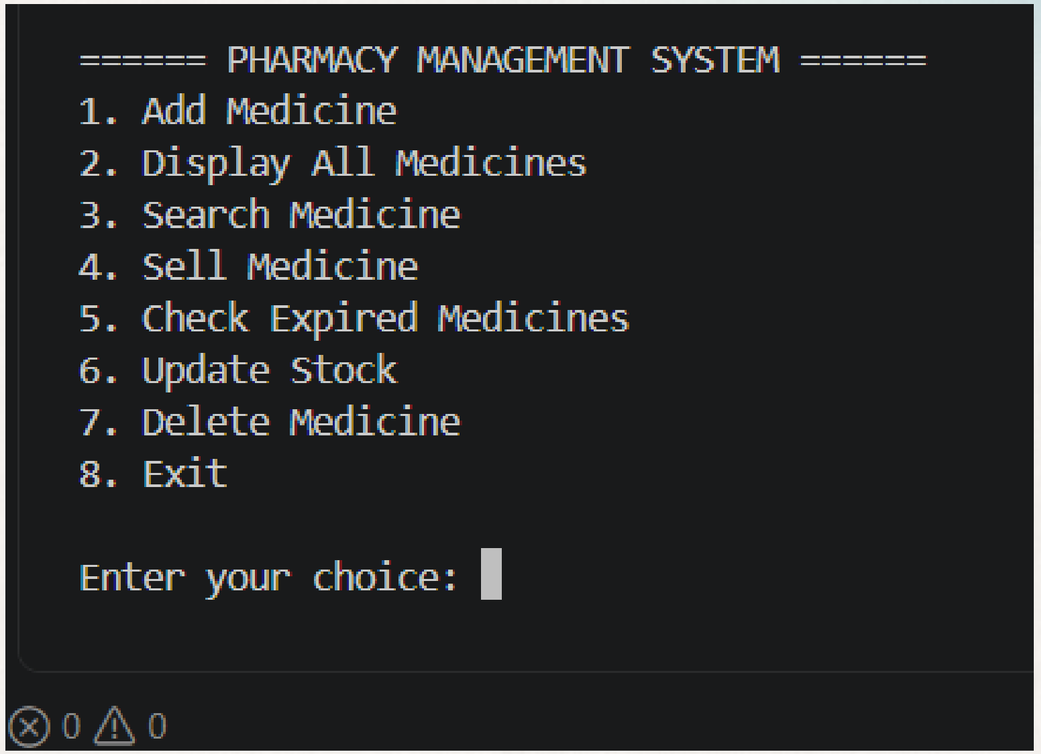
USER INTERFACE & KEY FEATURES

01

- Add with validation: No duplicate IDs or invalid dates
- Instant search: Find any medicine by ID
- Auto billing: Total = quantity × price

02

- Expiry report: Lists every expired item
- Stock control: Update or delete records



TESTING STRATEGY & TEST CASES

Method: black-box testing of every menu option.

7 / 7 test cases passed

01

Add valid medicine (ID 101, expiry 15/08/2027) → Medicine added successfully: Pass
Duplicate ID (ID 101 again) → Medicine ID already exists: Pass
Invalid date (Month = 13) → Invalid input: Pass

02

Sell expired medicine (Expiry 10/05/2025) → Cannot sell expired medicine: Pass
Over-selling (Sell 80, stock 50) → Insufficient stock: Pass

03

Valid sale (Sell 10 @ Rs. 2.5) → Bill Rs. 25, stock 40: Pass
Search missing ID (ID 999) → Medicine not found: Pass

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